

Daily Log: 2026/04/07

Recorder: Hardware Team

Task Done Today:

- Initiated chassis upgrade: Replaced legacy Dynamixel actuators with Feetech ST3215 servo motors.
- Designed and machined custom motor housings and mounting points to ensure compatibility with the new actuators.
- Conducted initial power-on and movement testing.

Issues and Solutions:

Issues	Solutions
Online 3D models of the Feetech motors did not match the motors' physical dimensions, resulting in misaligned mounting points.	Performed precision physical measurements of the actual hardware and created custom 3D models from scratch to ensure a perfect mechanical fit.

New Ideas / Planning:

- Proposed a significant redesign of the wheel assembly to allow for in-wheel motor integration.
- Planning to reorient the motor placement to a horizontal configuration, which will maximize internal chassis volume for the compute unit and battery arrays.

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Date: 2026/04/19

Recorder: Software Team

Task Done Today:

- Attempted to reinstall the OS, switching from Ubuntu 24.04 to Ubuntu 22.04. This was intended to improve compatibility with ROS2 Humble when controlling the ST3215 servos.

Issues and Solutions:

Issues	Solutions
System failed to boot continuously after installing Ubuntu 22.04.	Discovered that the Raspberry Pi 5 architecture only supports Ubuntu 23.04 and newer versions. Decided to reinstall Ubuntu 24.04 to ensure system functionality.

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Date: 2026/04/19

Recorder: Software Team

Task Done Today:

- Attempted to reinstall the OS, switching from Ubuntu 24.04 to Ubuntu 22.04. This was intended to improve compatibility with ROS2 Humble when controlling the ST3215 servos.

Issues and Solutions:

Issues	Solutions
System failed to boot continuously after installing Ubuntu 22.04.	Discovered that the Raspberry Pi 5 architecture only supports Ubuntu 23.04 and newer versions. Decided to reinstall Ubuntu 24.04 to ensure system functionality.

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Date: 2026/04/20

Recorder: Software Team

Task Done Today:

- Successfully completed the reinstallation of Ubuntu 24.04.
- Configured LiDAR, motor IDs, and the Python package management environment.

Issues and Solutions:

Issues	Solutions
LiDAR launch error: "Error, cannot bind to the specific [serial port:/dev/ttyUSB0] and [baudrate: 230400]"	The user lacked permissions for the dialout group. Resolved by executing <code>sudo usermod -aG dialout \$USER</code> and rebooting the system.
Modified motor IDs reverted to their original IDs after a power cycle.	The EEPROM must be unlocked prior to modification. Resolved by writing 0 to ADDR_LOCK (address 55) to allow permanent ID changes.
OS prevents pip installations to protect the system's Python environment.	Bypassed the restriction safely for development purposes by appending the <code>--break-system-packages</code> flag to pip install commands.

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Date: 2026/04/21

Recorder: Software Team

Task Done Today:

- Completed ST3215 servo control testing and integrated it with ROS2.
- Implemented a `/cmd_vel` topic subscriber to comply with the `slam_toolbox` control mechanism.
- Published the `/odom` topic to feed back the current linear and angular XYZ velocities.

- Integrated slam_toolbox and established the preliminary multi-terminal startup sequence (including ydlidar_ros2_driver, robot_state_publisher, st3215_base base_driver, /cmd_vel testing, slam_toolbox, and rviz2).

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Date: 2026/04/22

Recorder: Software Team

Task Done Today:

- Tested straight-line odometry (functions properly) but observed severe slipping during turns. Decided to integrate an IMU to correct the odometry.
- Successfully fused IMU data with wheel odometry using the robot_localization EKF node. Angular velocity readings are now highly accurate.
- Rerouted the slam_toolbox TF odometry source to be published by robot_localization instead of base_driver.
- Documented the IMU launch commands and the revised full system startup sequence.

Issues and Solutions:

Issues	Solutions
IMU linear_acceleration showed data, but angular_velocity remained consistently at 0.0.	Identified as a hardware failure. The issue was resolved by swapping in a new IMU module.
Map distortion at corners: When reaching a corner, the sudden disappearance of one side wall caused scan matching to pull the robot towards the remaining wall.	1. Reduced angle_variance_penalty from 1 to 0.5, making SLAM prioritize the EKF-fused odometry over scan matching. 2. Increased minimum_score to 0.7 to enforce stricter scan matching standards, preventing forced matches when similarities are low.

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Date: 2026/04/23

Recorder: Software Team

Task Done Today:

- Finalized and documented the complete 8-terminal startup sequence for the SLAM and Nav2 stack:
- LiDAR (ydlidar_launch.py)
- Base Driver (st3215_base base_driver)
- IMU (bno055)
- IMU Static TF (static_transform_publisher for base_link to bno055)
- Robot State Publisher (robot_state_publisher)
- EKF (robot_localization ekf_node)
- SLAM Toolbox (online_sync_launch.py)
- Nav2 (navigation_launch.py with nav2_slam_params_lattice.yaml)

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Date: 2026/04/26

Recorder: Software Team

Task Done Today:

- Tuned Nav2 parameters extensively, focusing on the costmap, inflation, and footprint.
- Adjusted the SmacPlannerLattice parameters to generate paths with wider turning angles (less smooth/rounded) to prevent the wheels from scraping the corners of the walls.
- Integrated frontier_explorer as Terminal 9 to automate goal generation.

New Ideas / Planning:

- Costmap Inflation: Set local costmap inflation to 0.1 (cost: 0.5) to keep paths away from walls. Set global costmap inflation to 0.092 (slightly wider than the robot) to prevent SmacPlannerLattice from hugging walls.
- Footprint Configuration: Set the global footprint to a smaller 16x16 cm to prevent "start occupied" or "no valid path found" errors. Set the local footprint to 18.2 x 18.2 cm to ensure safe clearances during turns.

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Date: 2026/05/09

Recorder: Software Team

Task Done Today:

- Documented systemd service commands for managing automated boot launches.
- Note for disabling boot services:
- `sudo systemctl stop robot-bringup.service`
- `sudo systemctl disable robot-bringup.service`

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Date: 2026/05/10

Recorder: Software Team

Task Done Today:

- Conducted a major path planning architecture update to resolve persistent wall-collision issues during turns.
- Displayed the footprint in RViz to enhance spatial awareness and improve debugging capabilities by accurately visualizing the robot's perceived position and orientation on the map.
- Updated Nav2 startup sequence (Terminal 8) to use `nav2_slam_params_2d.yaml`.

Issues and Solutions:

Issues	Solutions
SmacPlannerLattice fails in narrow maze environments (clearance as small as 5 cm). High inflation causes "no valid path" or "start occupied" errors, while low inflation causes wall collisions during turns.	Concluded that the Lattice planner is unsuited for this specific maze. Switched to SmacPlanner2D (A*-based), which is less complex. Increased the inflation radius to 1.2 to effectively center the path in the corridors, drastically reducing collisions. Enabled <code>allow_start_in_occupied: true</code> in SmacPlanner2D to allow successful path

	generation even if the robot's starting position falls within a heavily inflated zone.
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Daily Log: 2026/06/04

Recorder: Hardware Team

Task Done Today:

- Conducted field testing of the dual-Pi chassis (22x18 cm) within the maze environment.

Issues and Solutions:

Issues	Solutions
Large chassis dimensions (22 x 18 cm) led to poor maneuverability, limited fault tolerance during turns, and frequent collisions with speed bumps.	Determined that the chassis is physically too large for the maze constraints. Initiated the architectural decision to switch to the Jetson Orin NX platform for higher compute density and a more compact form factor.

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Daily Log: 2026/06/05

Recorder: Hardware Team

Task Done Today:

- Inspected the reComputer mini J4012 (Jetson Orin NX 16GB) package.
- Discovered structural instability in the original two-layer mounting standoff and screw design.
- Took precision measurements of components to determine necessary hardware upgrades.

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Daily Log: 2026/06/06

Recorder: Hardware Team

Task Done Today:

- Purchased and installed upgraded heavy-duty mounting standoffs.
- Completed the chassis migration to the Jetson Orin NX main controller.
- Mapped all peripheral connections to ensure compatibility with the Orin architecture.

Issues and Solutions:

Issues	Solutions
The Jetson Orin NX footprint is larger than that of previous boards, blocking the LiDAR's field of view.	Reoriented the rear motors horizontally and created a custom central groove in the chassis. Placed the Orin NX within this groove (ports facing up) to clear the path for the LiDAR module.

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Date: 2026/06/07

Recorder: Vision Team

Task Done Today:

- Connected and initialized the Logitech USB Camera.
- Developed a real-time image capture pipeline.
- Applied Grayscale Conversion, Gaussian Blur, and Canny Edge Detection to extract Regions of Interest (ROIs) for target letter localization.
- Displayed the real-time camera feed alongside the processed edge frame for visual debugging.
- Redesigned wheel geometry to implement an "in-wheel motor" mounting configuration.
- Finalized purchase order for the PD-capable power bank.

Issues and Solutions:

Issues	Solutions
Camera ROIs were highly susceptible to background interference, leading to numerous non-target objects being detected as candidate regions.	Adjusted the Gaussian Blur parameters to smooth out noise and implemented bounding box area constraints to ignore insignificantly small objects.
An increased wheel radius from the motor redesign rendered the original silicone tire casings incompatible.	Implemented a segmented tire strategy: sliced the silicone casing into small segments and adhered them to the wheel side to maintain traction.

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Date: 2026/06/08

Recorder: Vision Team

Task Done Today:

- Built the AI training dataset specifically for Letter Victims.
- Wrote an automated data collection script and gathered image data for various target letters (Psi, Phi, Omega).
- Utilized Histogram of Oriented Gradients (HOG) for feature extraction and trained a Support Vector Machine (SVM) classifier.
- Trained the AI model and displayed the real-time inference results.
- Realigned the LiDAR/IMU module to the absolute center of the chassis.
- Integrated the battery/compute layout into the central groove design.
- Verified the power bank model for PD logic supply.

Issues and Solutions:

Issues	Solutions
High false-positive rate: The model frequently misclassified random background objects as target letters.	Introduced an explicit "Unknown" category into the training dataset to help the SVM classifier filter out non-letter features. This

	effectively resolved the background misclassification issue.
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Date: 2026/06/09

Recorder: Vision Team

Task Done Today:

- Implemented Confidence Scores, Margin Thresholds, and Unknown Rejection mechanisms to reduce false positives and increase accuracy.
- Added programmatic logic to distinguish between Target Victims (visual targets) and Letter Victims.
- Fabricated the second iteration of the Jetson Orin NX-based chassis.
- Performed structural load testing on the new mounting points.

New Ideas / Planning:

- Key Finding: We discovered that training the AI model and performing inference on Edge Frames is significantly more stable and robust than using raw RGB images.

Issues and Solutions:

Issues	Solutions
The probability of letter misclassification was still relatively high. The system also tended to misclassify arbitrary curved objects as perfect circles.	Assigned similarity scores to detected letters. If the confidence score falls below a specific threshold, it is automatically rejected and classified as "Unknown." By tuning the decision thresholds, ambiguous targets are now strictly categorized as "Unknown," significantly improving overall accuracy.

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Date: 2026/06/10

Recorder: Vision Team

Task Done Today:

- Utilized contour analysis to identify circular candidate targets.
- Implemented a circle-validation algorithm: Divided the detected circles into four quadrants and verified the circular similarity of each segment. This effectively eliminates incomplete circles or shapes that merely resemble circles.
- Established a secondary ROI verification pipeline.
- Conducted final integration of the Orin NX control stack and power distribution.
- Verified communication between the Orin NX and peripheral modules.

Issues and Solutions:

Issues	Solutions
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Hardware transition challenges: Moving to the OpenMV H7 Plus camera introduced significant differences in syntax and system environment (MicroPython). Furthermore, the OpenMV lacks the RAM/computation power to run the SVM/AI model directly on board.	1. Utilize OpenMV's native libraries to extract ROIs (similar to the previous pipeline) and transmit only the cropped frames to the Raspberry Pi for heavy AI inference. 2. Process the simpler Target Victim detection entirely on the OpenMV H7 Plus and send only the final results to the Pi to reduce bandwidth.
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Date: 2026/06/11

Recorder: Hardware Team

Task Done Today:

- Finalized chassis geometry: Modified side walls to be solid for structural integrity and added chamfers to edges.
- Completed assembly of the Jetson Orin NX-based vehicle.
- Conducted final 3D printing run for updated wheels, main chassis, and camera housing.
- Implemented software-side vision pipeline improvements to support the hardware changes.

Issues and Solutions:

Issues	Solutions
Camera ROI is highly susceptible to background interference, leading to false-positive detections.	1. Adjusted Gaussian Blur parameters to filter high-frequency sensor noise. 2. Implemented area-based ROI filtering to ignore small background objects that do not meet the expected victim size profile. and send only the final results to the Pi to reduce bandwidth.

New Ideas / Planning:

- Conduct long-range stress testing of the new in-wheel motor mounts under full load.
- Verify the vibration dampening effect of the new 3D-printed chassis walls.

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Hardware Development Log: 2026 Season

Phase 1: Domestic Competition Vehicle (Based on Legacy Chassis)

Dates: 2026/04/18 – 04/26

Tasks: Modified legacy chassis to accommodate Raspberry Pi 5 + UPS. System configuration limited to essential modules: ST3215 motors, motor drivers, LiDAR, and IMU.

Focus: Core system stability and basic autonomous navigation verification for the local competition.

Phase 2: Dual-Pi Chassis Iteration

Dates: 2026/05/05 – 06/04

Tasks: Designed and printed a new chassis (22x18 cm) to house dual Raspberry Pi units.

Key Discovery (06/04): The dual-Pi chassis dimensions (22x18 cm) were found to be excessively large for efficient navigation in the confined maze. This limitation led to the architectural decision to switch to the Jetson Orin platform for higher compute-density.

Phase 3: Jetson Orin NX Integration

Dates: 2026/06/06 – 06/11

Tasks:

06/06: First chassis iteration for the Jetson Orin NX.

06/07: Mechanical redesign to implement "In-Wheel" motor mounting, significantly optimizing internal space for battery and compute placement.

06/09: Second chassis iteration.

06/11: Finalized chassis design, including solid side walls and chamfered edges to prevent snagging on maze walls.